Parametric Equations & Modeling

Section 1: Evaluating Parametric Equations and Graph Points

1 A particle moves according to $x(t) = t^2 - 3$ and $y(t) = 2t + 1$. What is its position at $t = 3$?

a. $(6, 7)$

b. $(9, 7)$

c. $(6, 6)$

d. $(3, 7)$

2 At what value of t does the parametric curve $x(t) = t^2 - 4$, $y(t) = t - 1$ intersect the y-axis with $y > 0$?

a. $t = 2$

b. $t = -2$

c. $t = 4$

d. $t = 1$

3 Given $x(t) = 4\cos t$ and $y(t) = 4\sin t$ for $0 \le t \le 2\pi$, what curve is traced and in which direction?

a. A circle of radius 4 traced counterclockwise

b. A circle of radius 4 traced clockwise

c. An ellipse of semi-major axis 4

d. A line segment from $(-4, -4)$ to $(4, 4)$

Section 2: Eliminating the Parameter

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4 Eliminate the parameter t from $x(t) = 3t + 2$ and $y(t) = 6t - 5$.

a. $y = 2x - 9$

b. $y = 2x - 7$

c. $y = 2x + 1$

d. $y = 3x - 11$

5 Eliminate the parameter from $x = 3\cos\theta$ and $y = 5\sin\theta$.

a. $\frac{x^2}{9} + \frac{y^2}{25} = 1$

b. $\frac{x^2}{25} + \frac{y^2}{9} = 1$

c. $x^2 + y^2 = 34$

d. $9x^2 + 25y^2 = 1$

6 Eliminate the parameter from $x = t^2$ and $y = t^4 + 3$ for $t \in (-\infty, \infty)$. What is the resulting Cartesian equation and its domain?

a. $y = x^2 + 3$ with domain $x \geq 0$

b. $y = x^2 + 3$ with domain $(-\infty, \infty)$

c. $y = 2x + 3$ with domain $x \geq 0$

d. $y = x^4 + 3$ with domain $x \geq 0$

Section 3: Projectile Motion Modeling

7 A projectile is launched with initial speed $v_0 = 80$ ft/s at an angle of 30° from the horizontal from ground level. Which parametric equations model its path ($g = 32$ ft/s^2)?

a. $x(t) = 40\sqrt{3}t$ and $y(t) = 40t - 16t^2$

b. $x(t) = 40t$ and $y(t) = 40\sqrt{3}t - 16t^2$

c. $x(t) = 80t$ and $y(t) = 80t - 32t^2$

d. $x(t) = 40\sqrt{3}t$ and $y(t) = 40t - 32t^2$

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8 For the projectile above with $y(t) = 40t - 16t^2$, what is its total flight time before landing?

a. 2.5 seconds

b. 1.25 seconds

c. 5.0 seconds

d. 3.0 seconds

9 What is the maximum height attained by a projectile modeled by $y(t) = 64t - 16t^2$?

a. 64 ft

b. 32 ft

c. 128 ft

d. 48 ft

Section 4: Direction of Motion and Speed

10 As parameter t increases from 0 to π, how does the curve $x = \sin t$, $y = \cos t$ move?

a. Clockwise from $(0, 1)$ to $(0, -1)$

b. Counterclockwise from $(1, 0)$ to $(-1, 0)$

c. Clockwise from $(1, 0)$ to $(-1, 0)$

d. Counterclockwise from $(0, 1)$ to $(0, -1)$

11 If a curve is given by $x(t) = t + 1$ and $y(t) = t^2$, what is the average rate of change $\frac{\Delta y}{\Delta x}$ from $t = 1$ to $t = 3$?

a. 4

b. 8

c. 2

d. 6

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12 Which parametric curve represents a straight line segment with endpoints $(1, 2)$ and $(7, 14)$ for $0 \le t \le 1$?

a. $x(t) = 1 + 6t, y(t) = 2 + 12t$

b. $x(t) = 7t, y(t) = 14t$

c. $x(t) = 1 + 7t, y(t) = 2 + 14t$

d. $x(t) = 6t, y(t) = 12t$

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